

Effort estimation using UCP Approach

$$UCP = UUCP \times TCF \times ECF$$

UUCP – Unadjusted Use Case Points

TCF – Technical Complexity Factor

ECF – Environmental Complexity Factor

$$UUCP = UAW + UUCW$$

UAW – Unadjusted Actor Weight

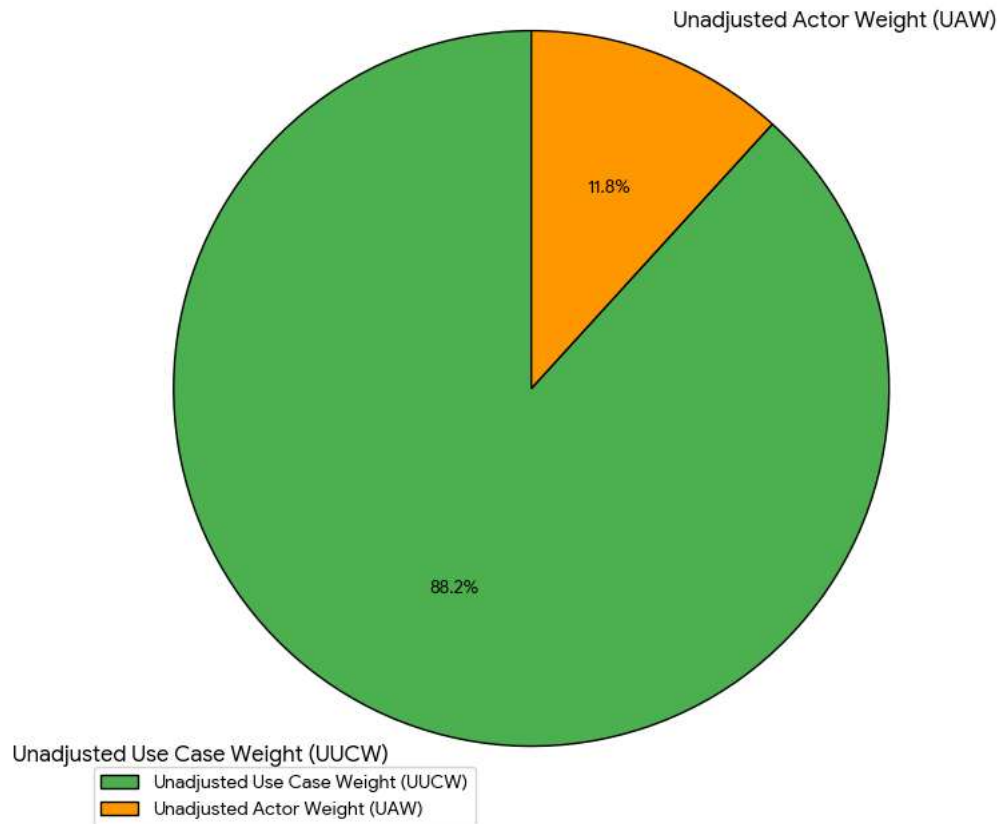
UUCW – Unadjusted Use Case Weight

Actor Type	Description	Count	Weight	Weighted Total
Simple (A1)	PRESTO System, Payment Service	2	1	$2 \times 1 = 2$
Average (A2)	(None assumed)	0	2	$0 \times 2 = 0$
Complex (A3)	Passenger, Maintenance Staff	2	3	$2 \times 3 = 6$
Total				UAW = 8

Use Case Type	Description	Transactions	Weight	Count	Weighted Total
Simple (UC1)	3 or less	3-	5	0	$0 \times 5 = 0$
Average (UC2)	4 to 7	4, 5	10	6	$6 \times 10 = 60$
Complex (UC3)	more than 7	8+	15	0	$0 \times 15 = 0$
Total					UUCW = 60

$$UUCP = UAW + UUCW = 8 + 60 = 68$$

Contribution to Unadjusted Use Case Points (UUCP)



The large slice representing UUCW confirms that the vast majority of the effort is driven by implementing the required transit functions (Buy Ticket, Recharge PRESTO, etc.), rather than managing complexity from external systems like the PRESTO or Payment Service stubs. This indicates a **Function-Centric** project.

$$TCF = C_1 + \left[C_2 \times \sum_{i=1}^{13} (W_{Ti} \times F_i) \right].$$

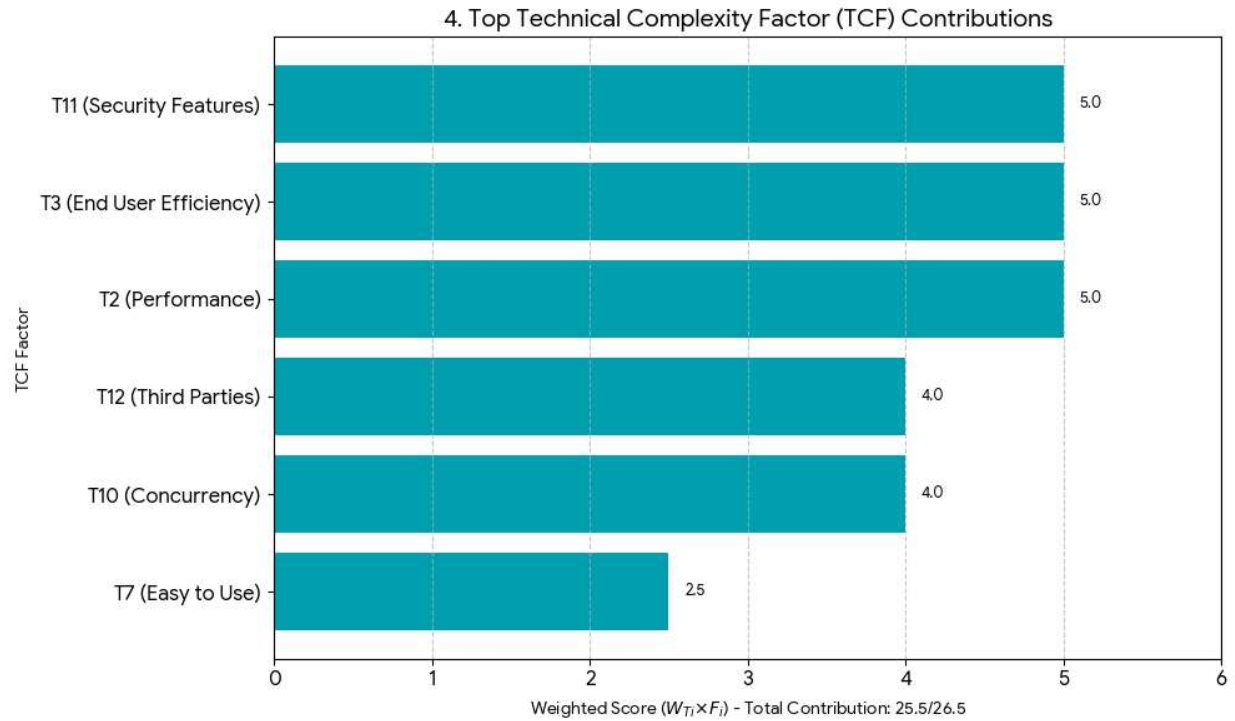
C1 = 0.6, C2 = 0.01

W_{Ti} = Technical Complexity Factor Weight

F_i = Perceived Impact Factor

TCF Type	Description	Weight (WTi)	Impact (Fi)	Weighted Score (WTi×Fi)
T2	Performance (<= 25s trans time)	1	5 (Strong Influence)	5.0
T3	End User Efficiency (Accessibility, ease of use)	1	5 (Strong Influence)	5.0
T11	Special Security Features (PCI-DSS, tokenization)	1	5 (Strong Influence)	5.0
T10	Concurrency (Processing during rush hour)	1	4 (Above Average)	4.0
T12	Provides Direct Access for Third Parties (Payment Gateway, PRESTO stub)	1	4 (Above Average)	4.0
T7	Easy to Use (Intuitive UI)	0.5	5 (Strong Influence)	2.5
<i>Other 7 Factors</i>	(Average / Low Influence)	(Sum of Weights: 7.5)	3 / 0	1.0 (Estimate)
Sum of Weighted Impact				26.5

$$TCF = 0.6 + [0.01 \times 26.5] = 0.6 + 0.265 = 0.865$$

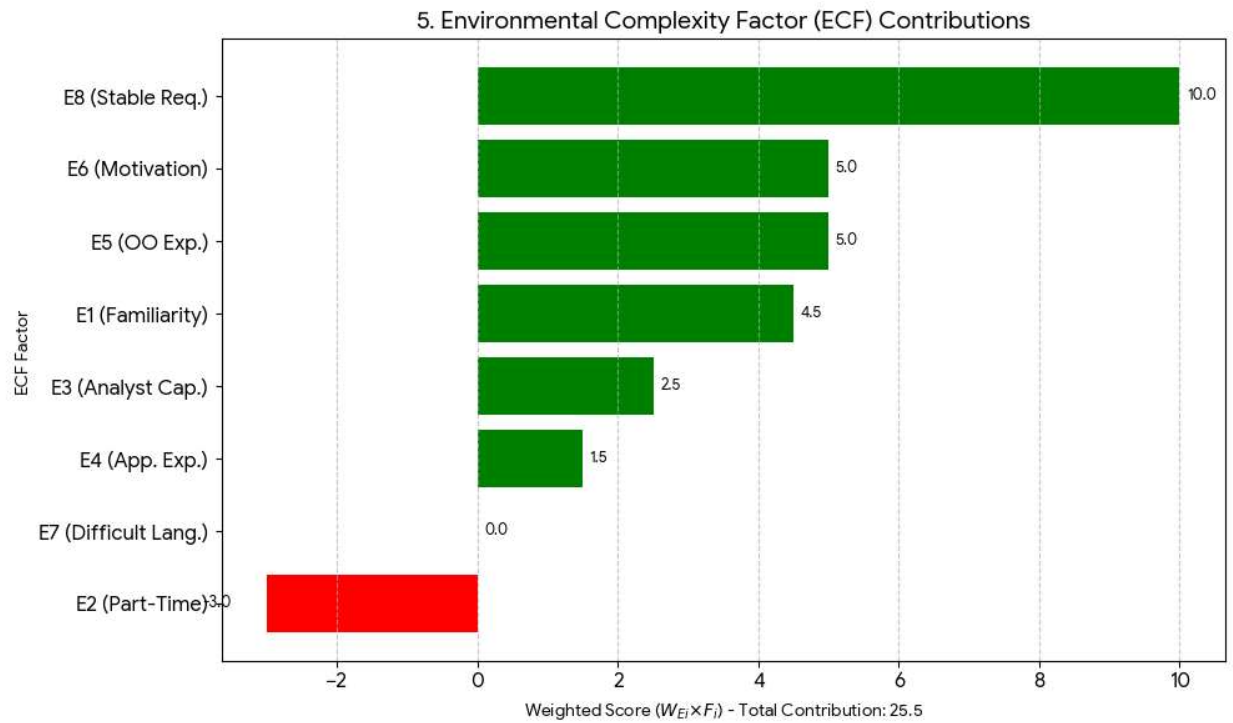


This confirms the technical complexity is driven by high scores in **Performance**, **End User Efficiency**, and **Security** (all 5.0), aligning perfectly with the core problem statement of improving a transit ticketing system.

$$ECF = C_1 + \left[C_2 \times \sum_{i=1}^8 (W_{Ei} \times F_i) \right].$$

ECF Type	Description	Weight (WEi)	Impact (Fi)	Weighted Score (WEi×Fi)
E8	Stable Requirements	2	5 (Strong, Favorable)	10.0
E3	Analyst Capability (GQM approach, detailed use cases)	0.5	5 (Strong, Favorable)	2.5
E5	Object-Oriented Experience (Implemented in Java, service-based)	1	5 (Strong, Favorable)	5.0
E6	Motivation	1	5 (Strong, Favorable)	5.0
E1	Familiarity with Use Case Domain	1.5	3 (Average)	4.5
E4	Application Experience	0.5	3 (Average)	1.5
E7	Difficult Programming Language (Java is standard)	-1	0 (No Influence)	0.0
E2	Part-Time Workers (Typically student project)	-1	3 (Average)	-3.0
Sum of Weighted Impact				25.5

$$ECF = 1.4 + [-0.03 \times 25.5] = 1.4 - 0.765 = 0.635$$



This shows the project benefited heavily from **Stable Requirements** (10.0 score) and the team's technical skills (**Object-Oriented Experience, Motivation**). It also clearly isolates the only major penalty, which is the **Part-Time Workers** factor (Score -3.0).

$$\text{UCP} = 68 \times 0.865 \times 0.635$$

$$\text{UCP} = 37.38$$

Effort Estimate = UCP x PF (Productivity Factor)

initial estimate for a new development team was 20 person-hours per UCP[Karner, 1993a; Karner, 1993b].

$$\text{Effort Estimate} = 37.38 \times 20$$

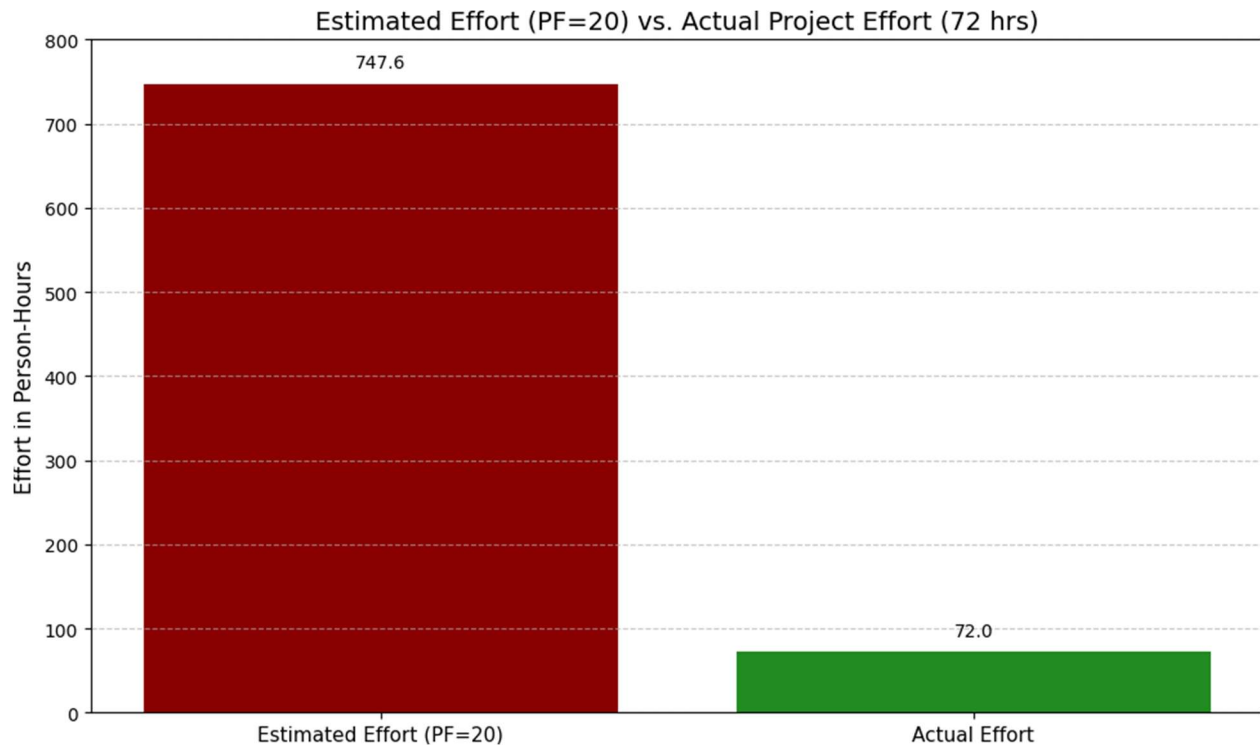
$$\text{Effort Estimate} = 747.6 \text{ Person-Hours}$$

Effort Distribution

Member	Role	Hours taken	Justification
Mohammad Naserameri	Manager	15	Oversaw project plan, led problem identification (D1) and integrated high-level metrics (UCP, SLOC) into D2. Focus on <i>planning and estimation</i>
Chen	Lead Back-end	14	Responsible for core logic like the PaymentService, SessionManager, and Transaction.java. High complexity due to handling financial logic and performance metrics.
Amishkumar	UI/UX	12	Focused on the front-end screens (e.g., MainMenuScreen, BuyTicketScreen) and ensuring bilingual and accessible UI implementation. Work includes FXML design

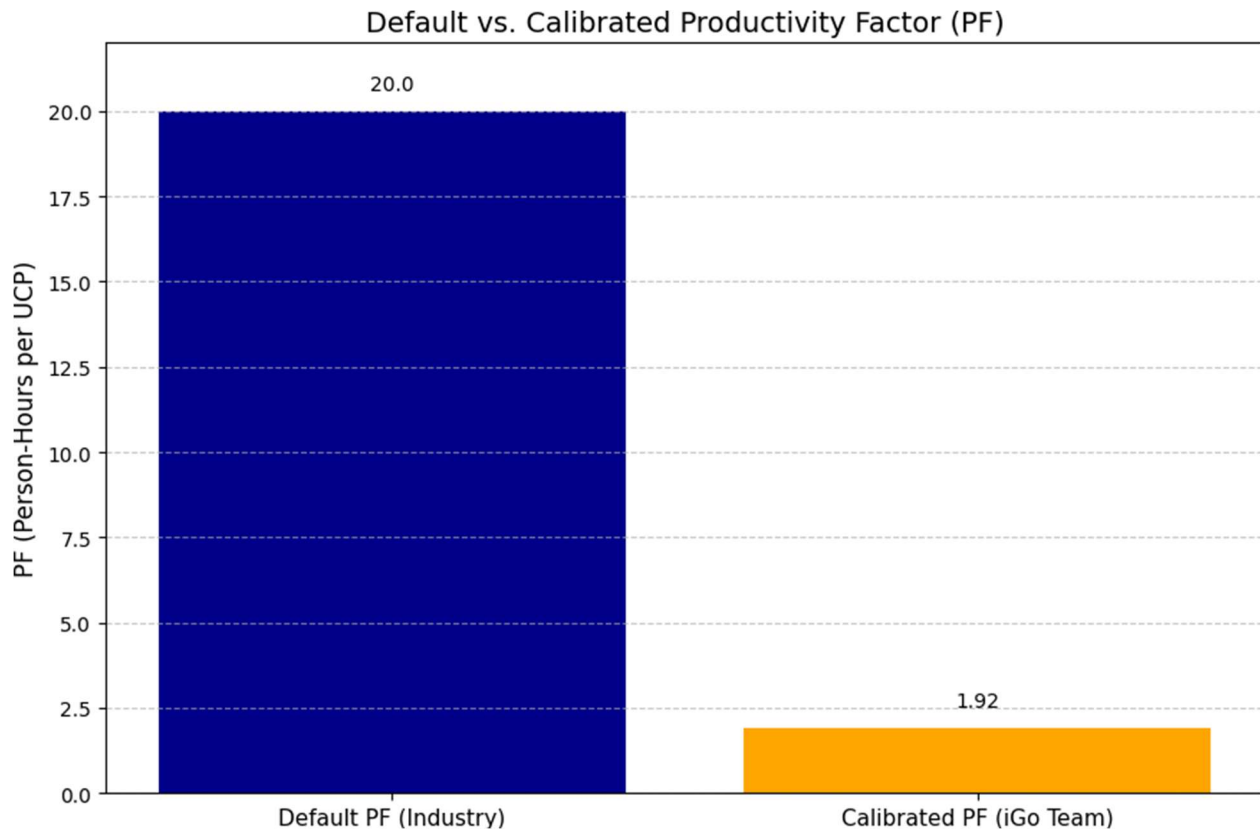
			and JavaFX Controller logic.
Diego	Testing	11	Primary responsibility for the performance simulation (SimulationDriver.java) and throughput validation (Q2). Also assisted with the SLOC/Comment Ratio analysis for D2
Poorav	Data Model	11	Focused on base system models (Transaction.java) and utilities like TokenUtil.java for data masking/security (Q9, Q11).
Ishmam	Documentation & Support	10	Responsible for organizing presentation content, referencing, finalizing use case textual descriptions (D1), and ensuring all D2 code was traceable to the repository.

Actual Effort = 72hrs



The towering red bar (747.6 hours) compared to the small green bar (72 hours) provides a powerful visual for the -938.33% deviation, emphasizing that the baseline industry model was wildly inaccurate for our project's context.

Calibrated PF = Actual Effort / UCP
Calibrated PF = 72 / 37.38
Calibrated PF = 1.92 hrs/UCP



This shows precisely *why* the effort deviated so severely. The Calibrated PF(1.92 hours/UCP) is the accurate measure of your academic team's productivity for this specific, limited-scope project, reflecting the "high efficiency and limited scope".

Comments:

Large Initial Deviation

The initial estimate using default industry standard PF=20 produced a massive deviation from the actual effort of 72hrs(approx. -90.5%)

Reason for Deviation:

The default PF=20 is suitable for complex production level, fully tested systems with standard labor hours. Our project was a time-boxed simulation of proof of concept using an experienced academic team.

This shows that UCP is useless without local calibration.

By calibrating the model with the actual effort, we find PF for our team is 1.92hrs/UCP.

This calibrated UCP accurately captures the high efficiency and limited scope(e.g., mock services, simple implementation, non-production quality) inherent in an academic project.